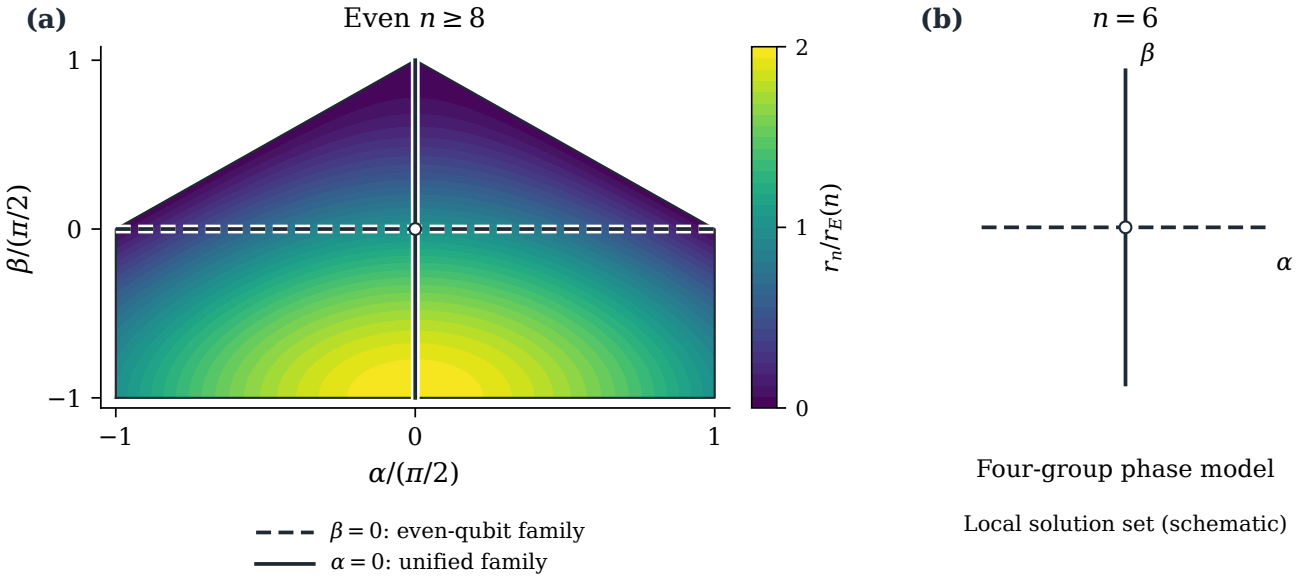

(a)
Even $n \geq 8$
1
0
−1
$\beta/(\pi/2)$
−1
0
1
$\alpha/(\pi/2)$
2
1
0
$r_n/r_E(n)$
$\beta = 0$: even-qubit family
$\alpha = 0$: unified family
(b)
$n = 6$
β
α
Four-group phase model
Local solution set (schematic)